

RELATIONSHIP BETWEEN MITOCHONDRIA AND OXYGEN CONSUMPTION IN ISOLATED CAT MUSCLES

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SUMMARY

1. Oxygen consumption, mitochondrial content and composition, intracellular lipid stores and fibre size were studied in isolated cat muscles: predominantly glycolytic gracilis, purely oxidative soleus and gracilis transformed into an oxidative muscle by chronic low-frequency (10 Hz) electrical stimulation.

2. Oxygen consumption in control gracilis at rest (0.303 ± 0.050 ml O_2 min⁻¹ 100 g⁻¹; mean $\pm$ S.E. of mean) was three to five times lower than in either stimulated gracilis (1.16 ± 0.40) or soleus (1.57 ± 0.56); it was about two times lower during maximal contractions in control gracilis (5.15 ± 0.24) than in either stimulated gracilis (11.6 ± 2.0) or soleus (9.34 ± 0.78).

3. The volume density of mitochondria in control gracilis ($2.75 \pm 0.12\%$) was half that of soleus (6.23 ± 0.76) and only one-third that of stimulated gracilis (8.35 ± 0.71). Subsarcolemmal mitochondria represented a significantly smaller fraction of the total mitochondrial volume in control gracilis than in either soleus or stimulated gracilis.

4. The surface area of inner and outer mitochondrial membranes per unit volume of mitochondria ranged from 23.4 to 26.1 and from 14.0 to 16.5 m² cm⁻³, respectively. Mean values of these variables were not significantly different among experimental groups.

5. The volume density of the intracellular lipid stores in control gracilis ($0.232 \pm 0.041\%$) was one-fourth of that in stimulated gracilis (0.860 ± 0.12) and one-fifth of that in soleus (1.17 ± 0.27).

6. The fibre cross-sectional area was 1670 ± 260 μm^2 in control gracilis, 2250 ± 280 in stimulated gracilis and 2390 ± 110 in soleus. The difference was statistically significant only between control gracilis and soleus.

7. There was a significant correlation between the volume density of mitochondria and maximal oxygen consumption for all three muscles combined.

8. It was found that mitochondrial structure was similar in muscles with different oxidative capacities and that equal amounts of mitochondria consumed equal amounts of oxygen under limiting conditions of maximal *in vivo* respiration.

INTRODUCTION

The relationship between maximal oxygen uptake ($\dot{V}_{O_2, \max}$) and various measures of oxidative capacity of skeletal muscle tissue has been studied extensively using endurance exercise as an experimental model (for recent reviews see Saltin & Gollnick, 1983; Holloszy & Coyle, 1984). The relative increase in total $\dot{V}_{O_2, \max}$ caused by endurance training is generally reported to be much smaller than the relative elevation of the oxidative capacity of the trained muscle tissue. This is found when muscle oxidative capacity is estimated by measuring activities of oxidative enzymes (Henriksson & Reitman, 1977; Davies, Packer & Brooks, 1981; Sjøgaard, 1984) as well as when muscle oxidative capacity is estimated morphometrically by measuring mitochondrial volume (Hoppeler, Howald, Conley, Lindstedt, Claassen, Vock & Weibel, 1985). Some reports, however, have shown a linear relationship between the volume density of mitochondria and $\dot{V}_{O_2, \max}$, when $\dot{V}_{O_2, \max}$ varied by as much as twofold between trained and untrained subjects (Hoppeler, Lüthi, Claassen, Weibel & Howald, 1973; Kiessling, Pilström, Bylund, Saltin & Piehl, 1974). Similarly, a comparison of animals of substantially different body masses revealed a constant, proportional relationship between $\dot{V}_{O_2, \max}$ and the total volume of mitochondria in three skeletal muscles for $\dot{V}_{O_2, \max}$ estimates differing by as much as fivefold (Mathieu, Krauer, Hoppeler, Gehr, Lindstedt, Alexander, Taylor & Weibel, 1981; Taylor, Maloiy, Weibel, Langman, Kamau, Seherman & Heglund, 1981). The interrelationship between whole-body $\dot{V}_{O_2, \max}$ and the oxidative capacity of skeletal muscle is thus currently a controversial issue (Saltin, 1986).

The correlation of functional parameters describing the global capacity of the respiratory system to carry oxygen with structural parameters characterizing the potential for oxygen transport and with consumption of a single muscle is far from ideal (Hoppeler, Lindstedt, Uhlmann, Niesel, Cruz-Orive & Weibel, 1984). The absolute amount of oxygen delivered to and consumed by any particular muscle selected for a structural or a biochemical analysis remains unknown. Hence, this approach may not be suitable for establishing causal relationships between functional variables of energy metabolism of skeletal muscle tissue and the structures which support them.

The present experiments were designed to estimate and correlate structure and function within a single muscle under limiting conditions of maximal oxygen transfer. Maximal blood flow and maximal oxygen consumption were studied in surgically isolated muscles which were later processed for an ultrastructural morphometric analysis. The study was carried out on a highly oxidative cat muscle (soleus) and a highly glycolytic muscle (gracilis; Ariano, Armstrong & Edgerton, 1973), as well as in gracilis muscle transformed from glycolytic to oxidative by chronic stimulation of the motor nerve at 10 Hz (Salmons & Vrbová, 1969; Hudlicka, Brown, Cotter, Smith & Vrbová, 1977).

Preliminary results of these experiments have been published previously in abstract form (Hoppeler, Hudlicka & Uhlmann, 1982, 1983).

METHODS

Experiments were performed on nineteen female cats of 2.2–4.6 kg body mass. In two cats in which control muscles were analysed we used the gracilis from one leg and the soleus muscle from the other leg.

Chronic electrical stimulation

Implantation of coiled stainless-steel electrodes for chronic stimulation was performed under halothane anaesthesia and aseptic conditions (Brown, Cotter, Hudlicka & Vrbová, 1976). The electrodes were sutured close to the entry of the nerve to the gracilis muscle, and led under the skin to the neck of the animals, where they were connected to a portable stimulator (Tyler & Wright, 1980) carried by the cats on a collar. Muscles were stimulated continuously for 8 h a day with rectangular pulses at a frequency of 10 Hz and of 0.3 ms duration. The voltage was adjusted to give contractions which could be felt and observed but did not cause obvious discomfort to the animals. Final experiments were performed after stimulation periods of 14–28 days.

Measurement of blood flow

Animals were anaesthetized with chloralose (70 mg/kg) and sodium pentobarbitone (Sagatal, 5 mg/kg) injected intraperitoneally. The trachea and jugular vein were cannulated. The anaesthesia was maintained with intravenous doses of Sagatal diluted with saline (1:1) as necessary. The blood supply to the soleus or gracilis muscle was isolated by tying off all vessels serving other muscles and tissues. Venous outflow was measured by a photo-electric drop counter (Devices Ltd, U.K.), connected to the femoral vein beyond its junction with the drainage of gracilis muscle with polyvinyl tubing which could be disconnected to insert glass capillaries for anaerobic withdrawal of blood samples. Blood was returned to the animal via the proximal part of the femoral vein. For soleus, the drop chamber was inserted between the popliteal and proximal femoral vein.

Drops were counted by an impulse integrator (Devices) and recorded on a pen recorder (Devices). The drop chambers were calibrated at the end of each experiment by measuring the volume of thirty drops of blood. Blood pressure was continuously recorded from the carotid artery by a pressure transducer (Bell & Howell Ltd, U.K.). The animals received heparin (1000 i.u./kg) prior to cannulation. The initial dose of heparin was supplemented every hour (500 i.u./kg). Muscles in which the blood supply could not be isolated because of the anatomical arrangement of the vascular supply were excluded from analysis.

Measurement of contractions

To measure the contractions of the soleus, the leg was fixed by drills inserted into the distal end of the femur and the tibia; for gracilis muscle the pubic bone and the distal end of the femur were fixed by clamps. The tendons were dissected free as far as possible, cut and connected to a strain gauge (Ether Ltd, U.K.), and isometric tension was recorded using a pen recorder (Devices). The broad and delicate distal tendon of gracilis did not allow an optimum recording of isometric twitch tension. Contractions were recorded during indirect stimulation of the peripheral end of the cut nerves close to their entry to either the gracilis or soleus muscle. All muscles were stimulated by supramaximal stimuli (1–3 V) of 0.3 ms duration activating all motor units. This was tested by gradually increasing the voltage until the tension could not be increased further by increasing the voltage. The length of the muscles was adjusted by means of a micrometer drive to obtain maximal muscle tensions. In a few pilot experiments isotonic contractions were used on the soleus muscle using a photo-electric cell and an optical wedge (Devices) which allowed recording of muscle shortening. In these experiments the muscle tendon was connected first to the optical wedge which was tied to a soft spring (allowing muscle shortening), connected to a strain gauge. In all experiments the muscle surface was kept moist and its temperature was maintained between 35 and 37 °C by a thermistor-controlled heating lamp.

Estimation of oxygen consumption

Oxygen consumption of the soleus muscle was measured in pilot experiments during both isotonic and isometric contractions using a variety of stimulation frequencies and loads. It was found that maximal oxygen consumption ($\dot{V}_{O_2, \max}$; protocol see below) was more reproducibly achieved with isometric than with isotonic contractions and consequently isometric contractions were used to

estimate $\dot{V}_{O_2, \max}$ in all experiments. Muscle oxygen consumption was calculated from blood flow and arterial-venous oxygen content differences. Oxygen content was measured in blood taken under anaerobic conditions from the carotid artery and the appropriate venous drainage. The capillary tubes containing the blood samples were clamped at both ends and stored on ice until analysed (not longer than $\frac{1}{2}$ h). A Lex-O₂-Con analyser (Lexington Instruments Ltd., U.S.A.) was used to measure oxygen content.

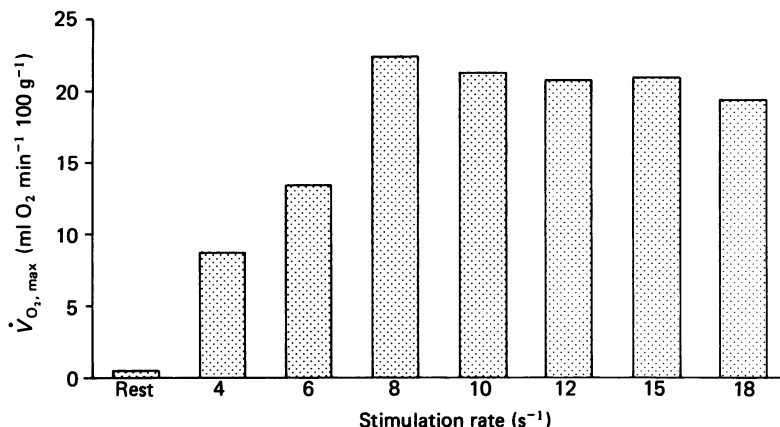


Fig. 1. Oxygen consumption rates at various stimulation rates for a gracilis muscle stimulated for 28 days. It was typically found that there was a sharp drop in oxygen consumption at stimulation rates lower than 8 Hz. In contrast, at higher stimulation rates (> 10 Hz) oxygen consumption declined only gradually.

Resting oxygen consumption was determined at the beginning of each experiment at least $\frac{1}{2}$ h after the completion of cannulation. $\dot{V}_{O_2, \max}$ was achieved in all muscles according to the following protocol: muscles were contracting for 120 s with an arterial and a venous blood sample being taken after 100 s. Oxygen consumption was first determined at a stimulation frequency of 8 Hz. The stimulation frequency was then increased to 10 Hz. If the oxygen consumption were higher at 10 than at 8 Hz, the stimulation frequency was further increased to 12 Hz. If on the other hand the oxygen consumption was lower at 10 than at 8 Hz, the stimulation frequency was lowered to 6 Hz. In most cases $\dot{V}_{O_2, \max}$ was either reached at 8 or 10 Hz (Fig. 1). The interval between individual bouts of stimulation was 20 min, by which time blood flow returned to pre-stimulation levels. The protocol described was established to minimize the number of trials necessary to determine peak oxygen consumption for each muscle preparation thereby limiting blood loss and potential ultrastructural damage to muscle fibres. Typically experiments took 6–7 h to complete.

Tissue sampling and processing

After the completion of the physiologic experiment muscles were dissected out, blotted with a saline-moistened gauze and weighed. The muscles were then cut into thin (1–2 mm wide) longitudinal strips, which were aligned in a linear sequence. From the aligned strips a total of six tissue samples were removed at random intervals. This procedure ensured that any location in a muscle had the same chance of being sampled. Tissue samples were fixed by immersion into a 6.25% (v/v) solution of glutaraldehyde in 0.1 M-sodium cacodylate buffer adjusted to 430 mosm with NaCl (total osmolality of the fixative was 1100 mosm, pH 7.4) and further processed for electron

Fig. 2. Cross-sections of portions of cat soleus (*A*), control gracilis (*B*) and chronically stimulated gracilis muscles (*C*). Mitochondria are more abundant in soleus and stimulated gracilis than in control gracilis. Note prominent lipid droplets in soleus (m.s. = subsarcolemmal mitochondria, m.c. = intermyofibrillar mitochondria, li. = lipid droplets, ca. = capillary, e.c. = erythrocyte, my. = myofibrils, nu. = nucleus, fi. = fibroblast, pe. = pericyte).

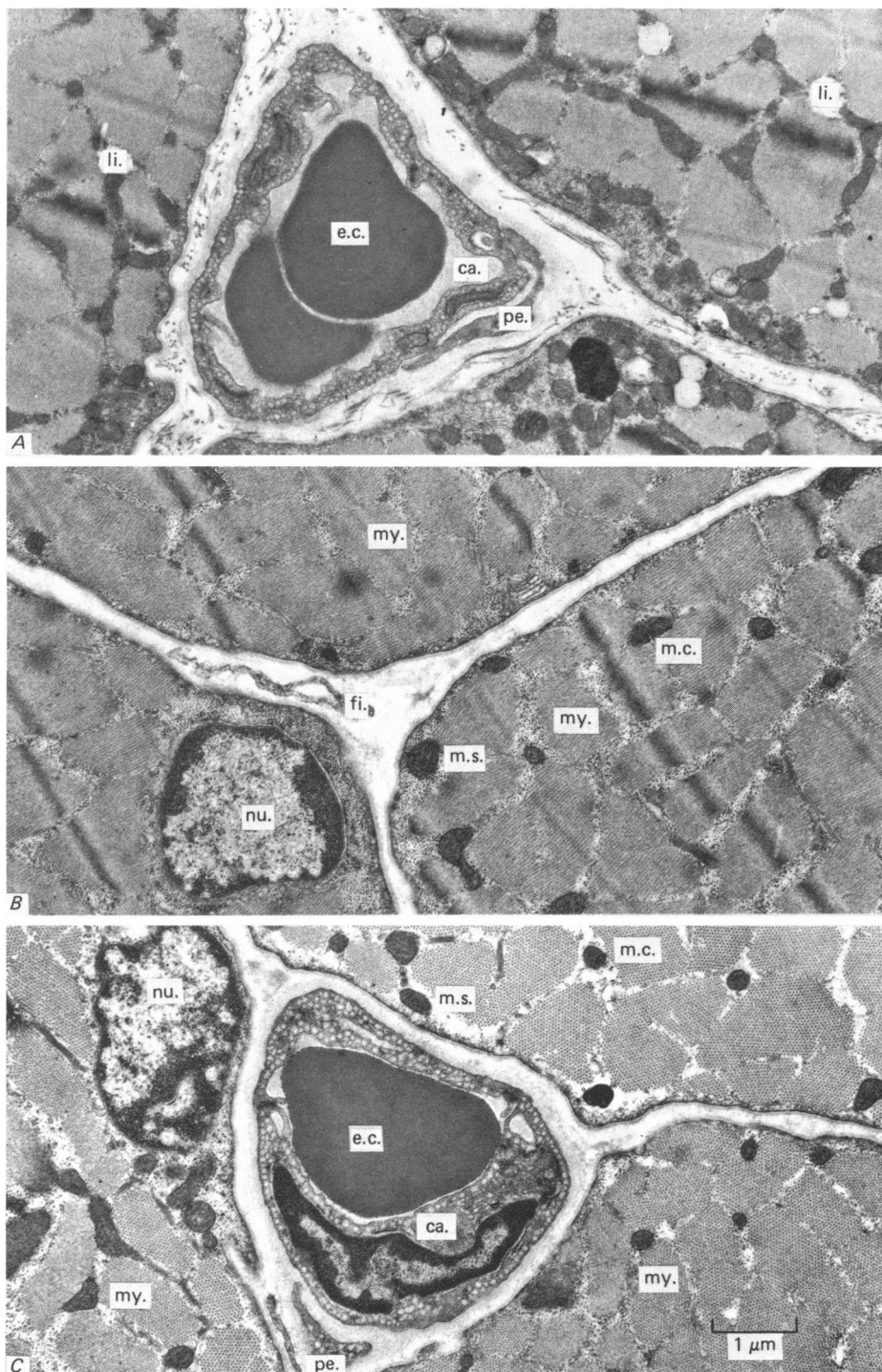


Fig. 2. For legend see opposite.

microscopy as described in detail by Hoppeler, Mathieu, Krauer, Claassen, Armstrong & Weibel (1981). To preserve the alignment of the muscle fibres for an analysis of capillary anisotropy (results not reported) some muscles were fixed *in situ* by vascular perfusion with the fixative mentioned above at a non-pulsatile pressure of less than 100 mmHg (Mathieu, Cruz-Orive, Weibel & Hoppeler, 1983). Muscles which were perfusion fixed were excluded from the analysis of fibre size because they were fixed at considerably longer sarcomere lengths, and hence smaller fibre diameters, than immersion-fixed muscle tissue.

Morphometry

For the morphometric analysis, four tissue samples per muscle were chosen at random and sectioned for electron microscopy. The orientation of the tissue sections was essentially transverse with regard to the fibre axis (Fig. 2). For the estimation of the fibre cross-sectional area, a final magnification of $1500\times$ was used. Four micrographs per section (and hence sixteen micrographs per muscle) were taken in consecutive frames of slotted grids, yielding for analysis about 150 muscle fibre profiles per muscle. A final magnification of $24\,000\times$ was used for the estimation of the volumes of mitochondria, lipid droplets and myofibrils per unit volume of muscle fibre. Ten micrographs were obtained from each section (and hence forty micrographs in each muscle) by systematically sampling from consecutive frames on 100-square mesh grids. All volume density estimates were thus obtained by sampling tissue from at least forty individual muscle fibres. The final magnification for the estimation of the surface of mitochondrial membranes per unit volume of mitochondria was $146\,000\times$. One section per muscle and four animals in each experimental group were analysed. Ten micrographs of intermyofibrillar and subsarcolemmal mitochondria were taken from different randomly selected muscle fibres in each muscle. All stereological variables were obtained by standard procedures described by Weibel (1979). Point counting was performed on a square lattice test system with a total of 100 points for the low, 144 points for the intermediate and 576 points for the high magnification.

To test the significance of differences between stimulated gracilis and control gracilis as well as between stimulated gracilis and control soleus. Student's *t* test for independent samples was used with the level of statistical significance set at 5%. Linear least-squares regression analysis (with the assumption of a normal distribution of errors) was used to describe the relationship between parameters.

RESULTS

Physiological data

Table 1 shows the averaged physiological data of the three experimental groups. The average body mass of the cats was close to 3 kg and not significantly different among groups. The average mass of gracilis muscles was almost identical in control and stimulated animals (6.97 ± 0.89 ; mean $\pm$ s.e. of mean and 7.04 ± 0.74 g respectively) with individual muscle masses ranging from 4.35 to 10.9 g. Soleus mass ranged from 1.65 to 3.51 g depending on animal size.

Oxygen extraction was between 20 and 35% at rest and did not differ significantly among the three experimental groups. At $\dot{V}_{O_{2, \max}}$, extraction was approximately doubled. There was no significant difference in extraction at $\dot{V}_{O_{2, \max}}$ between soleus and control gracilis muscles (60.7 and 64.3%, respectively) but a significantly higher extraction (77.5%) was observed in stimulated gracilis muscle (Table 1).

Soleus muscles and stimulated gracilis muscles had significantly higher resting and maximal oxygen consumptions than control gracilis muscles (Table 1). The highest value for oxygen consumption ($22.4 \text{ ml O}_2 \text{ min}^{-1} 100 \text{ g}^{-1}$) was recorded in a gracilis muscle stimulated for 28 days. This particular muscle also had the highest volume density of mitochondria of all muscles analysed (10.7%).

TABLE 1. Physiological characteristics of cat soleus and gracilis muscles. Means $\pm$ s.e. of mean; * denotes significantly different ($2P < 0.05$) from chronically stimulated gracilis muscles (14–28 days); † denotes $n = 4$; force is expressed in $N 10^{-2} g^{-1}$ muscle tissue

	Body mass (kg)	Muscle mass (g)	Peak force ($N g^{-1} 10^{-2}$)	Blood flow		Oxygen extraction		Oxygen consumption	
				Rest (ml min ⁻¹ 100 g ⁻¹)	Max. (ml min ⁻¹ 100 g ⁻¹)	Rest (%)	Max. (%)	Rest (ml O ₂ min ⁻¹ 100 g ⁻¹)	Max. (ml O ₂ min ⁻¹ 100 g ⁻¹)
Control soleus ($n = 8$)	2.84 $\pm$ 0.20	2.47 $\pm$ 0.23*	239 $\pm$ 29*	33.3 $\pm$ 11*	95.1 $\pm$ 13	32.1 $\pm$ 6.5	60.7 $\pm$ 5.0*	1.57 $\pm$ 0.56	9.34 $\pm$ 0.78
Control gracilis ($n = 6$)	2.73 $\pm$ 0.20	6.97 $\pm$ 0.89	33.9 $\pm$ 8.7†	8.43 $\pm$ 1.8*	44.5 $\pm$ 3.5*	22.4 $\pm$ 5.4	64.3 $\pm$ 5.2	0.303 $\pm$ 0.050*	5.15 $\pm$ 0.24*
Stimulated gracilis ($n = 7$)	3.39 $\pm$ 0.31	7.04 $\pm$ 0.74	43.3 $\pm$ 4.3	15.5 $\pm$ 2.6	87.3 $\pm$ 10	33.5 $\pm$ 6.7	77.5 $\pm$ 5.1	1.16 $\pm$ 0.40	11.6 $\pm$ 2.0

TABLE 2. Volume density of total mitochondria, interfibrillar mitochondria, subsarcolemmal mitochondria, intracellular lipid deposits and of myofibrils, and mean fibre cross-sectional area of cat soleus and gracilis muscles. Means $\pm$ s.e. of mean; * denotes significantly different ($2P < 0.05$) from chronically stimulated gracilis (14–28 days); † denotes $n = 4$; ‡ denotes $n = 5$; †† denotes $n = 7$

	Volume density					Mean fibre cross-sectional area (m ²)
	Total mitochondria (%)	Interfibrillar mitochondria (%)	Subsarcolemmal mitochondria (%)	Intracellular lipid deposits (%)	Myofibrils (%)	
Control soleus ($n = 8$)	6.23 $\pm$ 0.76	5.72 $\pm$ 0.76	0.504 $\pm$ 0.13	1.17 $\pm$ 0.27	81.6 $\pm$ 1.2*	2390 $\pm$ 110††
Control gracilis ($n = 6$)	2.75 $\pm$ 0.12*	2.59 $\pm$ 0.11*	0.171 $\pm$ 0.051*	0.232 $\pm$ 0.041*	81.6 $\pm$ 0.98*	1670 $\pm$ 260†
Stimulated gracilis ($n = 7$)	8.35 $\pm$ 0.71	7.38 $\pm$ 0.69	0.969 $\pm$ 0.19	0.860 $\pm$ 0.12	75.1 $\pm$ 2.5	2250 $\pm$ 280††

Structural data

Fibre cross-sectional area. The mean fibre cross-sectional area was not significantly different between control and stimulated gracilis muscles (Table 2). As the masses of these muscles were also not significantly different, we concluded that the total fibre number did not change with stimulation.

The volume density of myofibrils. The volume density of myofibrils was identical in soleus and control gracilis muscle (81.6%). In stimulated gracilis muscle the myofibrillar volume was reduced significantly to 75.1%.

Mitochondrial volumes. The data for the volume density of intermyofibrillar, subsarcolemmal and total mitochondria are reported in Table 2. It had previously been shown that the two populations of mitochondria adapt differently to stimuli such as endurance training (Hoppeler *et al.* 1985). All control gracilis muscles showed similar values for the volume density of total mitochondria (range, 2.41 to 3.11%). In contrast, mitochondrial densities in soleus covered a range from 3.18 to 9.43% and in stimulated gracilis from 5.43 to 10.7%. There was no systematic relationship between the duration of stimulation and the mitochondrial density. This may be attributed to the difficulty in obtaining uniform levels of electrical stimulation in all animals.

Membrane areas in mitochondria. Table 3 shows the data for the surface area of inner and outer mitochondrial membranes per unit volume of mitochondria. The differences between experimental groups and between subsarcolemmal and intermyofibrillar mitochondria were not statistically significant.

Assuming the whole muscle to be myofibres, the absolute volume of mitochondria in a muscle can be calculated as the product of the volume density of mitochondria and the muscle volume (the muscle volume being obtained by dividing the muscle mass by its specific weight of 1.06 g cm^{-3} ; Mendez & Keys, 1960). Assuming the whole-muscle mass to be fibres is obviously not correct as the interstitial space is thus neglected. This space is often widened in immersion-fixed skeletal muscle tissue (Fig. 2) because muscle fibres tend to segregate during tissue processing. This renders the correct morphometric analysis of that compartment difficult. Olivetti, Anversa & Loud (1980) reported the interstitial space of optimally perfusion-fixed rat hearts to be 15% of the total muscle volume, half the space being taken up by capillaries, the remainder by connective tissue elements. Since rat heart muscle has much smaller muscle fibres and a much higher capillary density than cat skeletal muscles it can be assumed that in the latter the interstitial space accounts for less than 10% of the total muscle volume.

Knowing the absolute volume of mitochondria in a muscle permits the calculation of the absolute quantities of oxygen consumed per unit volume of mitochondria and per unit time for that muscle. Mitochondrial oxygen consumption (units: $\text{ml O}_2 \text{ min}^{-1} \text{ cm}^{-3}$) ranged from 1.12 to 1.92 in soleus, from 1.29 to 2.29 in control gracilis and from 0.81 to 1.97 in stimulated gracilis muscles. As we estimated the surface areas of mitochondrial membranes we could calculate oxygen flux per unit area of outer and inner mitochondrial membranes (units: $\text{ml O}_2 \text{ min}^{-1} \text{ m}^{-2}$). For all muscles analysed the values ranged from 0.0579 to 0.142 for outer mitochondrial membranes and from 0.0328 to 0.0910 for inner mitochondrial membranes. A highly

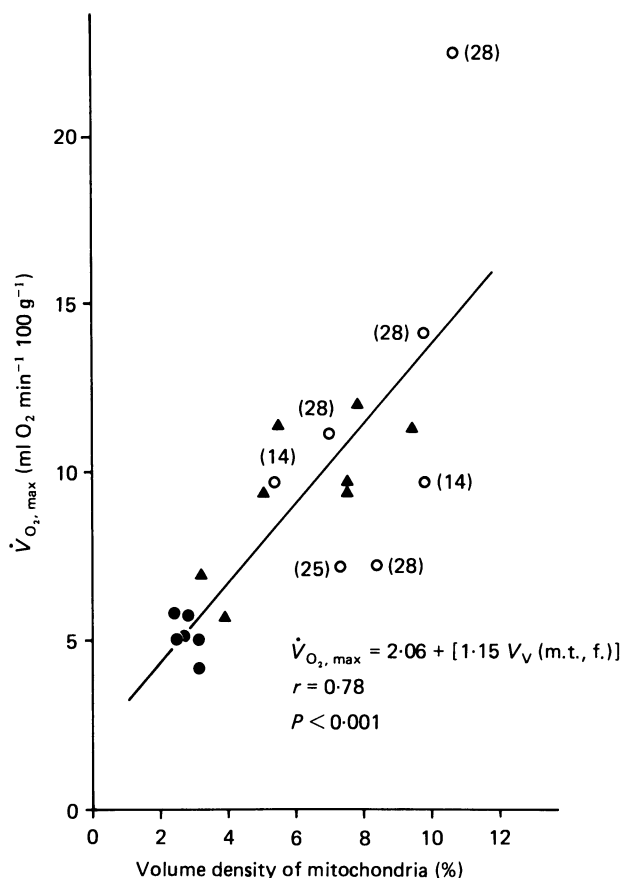


Fig. 3. Relationship between the volume density of mitochondria (V_v (m.t., f)) and the maximal oxygen consumption in isolated cat muscles. ●, control gracilis; ▲, soleus; ○, stimulated gracilis; numbers in parentheses indicate duration of electrical stimulation in days.

TABLE 3. Surface area of inner and outer mitochondrial membranes per unit volume of mitochondria of a subsample of cat soleus and gracilis muscles (means $\pm$ S.E. of mean; units, $\text{m}^2 \text{cm}^{-3}$)

	Intermyofibrillar mitochondria		Subsarcolemmal mitochondria	
	Inner membranes	Outer membranes	Inner membranes	Outer membranes
Control soleus ($n = 4$)	24.3 ± 2.1	16.5 ± 1.6	23.8 ± 0.78	15.9 ± 1.8
Control gracilis ($n = 4$)	25.2 ± 1.3	16.1 ± 1.0	23.4 ± 1.5	15.0 ± 1.0
Stimulated gracilis ($n = 4$)	24.7 ± 1.8	14.0 ± 1.1	26.1 ± 1.4	14.3 ± 0.90

significant linear relationship between the volume density of mitochondria and the maximal oxygen consumption was found when these variables were analysed for the three experimental groups combined (Fig. 3). It may be noted that the amount of tension that could be recorded during contractions of soleus muscles was much larger than during contractions of gracilis muscles (Table 1). This was probably mostly due to problems of attaching the gracilis tendons to the transducer. The dissection of the muscle for the structural analysis indicated that both muscles had a similar low degree of pennation. The fact that soleus and gracilis muscles showed a nearly identical relationship between the volume density of mitochondria and $\dot{V}_{O_{2, \max}}$ suggests that externally produced tension was not an important determinant of oxygen consumption under the experimental conditions chosen.

DISCUSSION

Mitochondrial membranes

The outer mitochondrial membrane represents the final diffusion barrier for oxygen before reaching the mitochondrial sink, while the inner mitochondrial membrane contains the respiratory chain units and thus establishes a transduction for energy from oxygen to ATP (Weibel, Taylor, Gehr, Hoppeler, Mathieu & Maloiy, 1981). The present study shows that the amount of either mitochondrial membrane area per unit volume of mitochondria is not significantly different between highly oxidative and highly glycolytic muscles, and not affected by chronic electrical stimulation (Table 3). These findings are consistent with the data reported for mitochondria in chronically stimulated rabbit tibialis anterior muscle (Reichmann, Hoppeler, Mathieu-Costello, von Bergen & Pette, 1985) as well as in strenuous endurance exercise (Hoppeler *et al.* 1973; Davies *et al.* 1981). Since the surface area of mitochondrial membranes per unit volume of mitochondria remained constant in the present experiment, the mitochondrial volume is an adequate descriptor of the muscles' potential for aerobic energy conversion (Schwerzmann, Cruz-Orive, Eggmann, Sanger & Weibel, 1986).

Volume densities of mitochondria

In soleus muscle the volume density of mitochondria varied by as much as threefold between individual animals (Fig. 3). This was a surprising finding as soleus muscle in cat is presumed to be composed of slow-twitch oxidative fibres only (Ariano *et al.* 1973). We therefore expected mitochondrial densities of muscle fibres to be rather constant not only within a muscle of a single animal but also between muscles from various animals. The homogeneity of the muscle fibre composition in our animals was checked by histochemistry on two soleus muscles with a largely different mitochondrial density using the ATPase method of Padykula & Herman (1955). In both cases more than 95% of all fibres were slow-twitch fibres. Mitochondrial densities in soleus muscle were neither related to the age nor to the weight of the animals and we could find no apparent reason for the large differences in the volume density of mitochondria of this muscle between individual animals.

Stimulated muscles have greatly increased activities of the major enzymes of the oxidative pathways of energy production (Pette, Smith, Staudte & Vrbová, 1973;

Brown *et al.* 1976; Heilig & Pette, 1980). One could therefore expect to find a larger mitochondrial compartment in stimulated than in control muscles. In the present study mitochondrial densities were increased by as much as fourfold over control values in some muscles stimulated for 28 days (Fig. 3). As the muscle mass was not affected by the stimulation procedure the change in the volume density of mitochondria directly reflects the change in the absolute volume of mitochondria in stimulated muscles (Mathieu *et al.* 1981). The changes in the volume densities of mitochondria agree with morphometric data on mitochondria in rabbit tibialis anterior muscle stimulated for similar periods of time (Eisenberg & Salmons, 1981; Eisenberg, Brown & Salmons, 1984). In agreement with Eisenberg & Salmons (1981), we found that subsarcolemmal mitochondria increased comparatively more with electrical stimulation than intermyofibrillar mitochondria (Table 2). No preferential adaptation of subsarcolemmal mitochondria was found, however, in rabbit tibialis anterior muscle (Reichmann *et al.* 1985) electrically stimulated for 2–28 days. The reason for this difference might be related to the use of an intermittent stimulation protocol (1 h on, 1 h off) in the latter study instead of the continuous 8 h stimulation used in the present and other experiments (Eisenberg & Salmons, 1981; Eisenberg *et al.* 1984).

Volume density of intracellular lipid deposits

The lowest volume density of intracellular lipids was found in control gracilis muscles. Only 0.2% of the fibre volume was occupied by lipid droplets. Chronic electrical stimulation significantly increased by fourfold the volume density of lipid droplets in gracilis muscles bringing values close to those observed in solei (1.2%; Table 2). Changes in intracellular lipid stores similar to those observed in stimulated gracilis have been described in endurance exercise (Hoppeler *et al.* 1973; Kiessling *et al.* 1974) as well as in adaptation to cold environment (Buser, Kopp, Gehr, Weibel & Hoppeler, 1982). The observed changes might be related to an increased importance of fat as a fuel for mitochondrial oxidation after adaptation to chronic stimulation (Hudlicka *et al.* 1977). No changes in intracellular lipid stores were found in a recent experiment on intermittently stimulated rabbit tibialis anterior muscle (Reichmann *et al.* 1985).

Oxygen consumption

Resting oxygen consumption of normal gracilis muscle was significantly lower than that of either soleus or stimulated gracilis muscles. As the oxygen extraction at rest was similar in all muscles (approximately 30%) this was due to the smaller resting blood flow in glycolytic as compared to oxidative muscles (Table 1). Our data are thus in agreement with most data reported in the literature (Folkow & Halicka, 1968; Hilton, Jeffries & Vrbová, 1970; Hudlicka, 1969, 1975; Petrofsky, Phillips, Sawka, Hanpeter & Stafford, 1981).

The stimulation protocol used to reach $\dot{V}_{O_2, \max}$ enabled us to determine peak oxygen flow with only three trials in most cases (see Methods section). Values for peak oxygen consumption and extraction in soleus and gracilis muscles measured in this study (Table 1) are very close to values reported for the same variables in cat soleus and gastrocnemius by Folkow & Halicka (1968) who used a similar stimulation

protocol. In contrast to these authors, we found no increase in $\dot{V}_{O_2, \max}$ when stimulating soleus muscles at frequencies higher than 12 Hz.

The major concern about the functional variables studied was to obtain a reliable estimate of the maximal potential for oxygen consumption of individual muscles. One might therefore ask the question whether $\dot{V}_{O_2, \max}$ measured in isolated muscles of an anaesthetized animal using trains of isometric twitches could come anywhere close to 'true' $\dot{V}_{O_2, \max}$ of the muscle when used by the conscious animal during maximal exercise. The mean arterial pressure, and hence the pressure available for the perfusion of the vascular bed, is considerably higher in an exercising than in an anaesthetized animal. It was shown that in resting muscle both oxygen consumption and blood flow are strongly dependent on perfusion pressure (Whalen, Buerk & Thuning, 1973). Furthermore, an acutely exercising animal undergoes profound hormonal adjustments which greatly enhance substrate availability to working muscles (Newsholme, 1977). We do not know the extent to which these factors influence oxygen consumption measurements done under limiting conditions on surgically isolated muscles and how much influence the anaesthetic agents *per se* might have on mitochondrial respiration.

Another important consideration is that during voluntary contractions, muscles are not activated by synchronous isometric twitches. This artificial mode of muscle excitation has in fact been shown to be energetically rather uneconomical. In rat soleus muscle the energy cost of tension development was found to be four times higher when tension was produced with unfused twitches instead of fully fused tetany achieved by asynchronous activation of motor units (Loiselle & Walmsley, 1982). Petrofsky *et al.* (1981) have analysed oxygen consumption of cat soleus and gastrocnemius during graded tetanic contractions obtained by sequential stimulation of ventral root filaments. This stimulation pattern is claimed to be very similar to the physiological asynchronous recruitment of motor units. Values for peak oxygen consumption observed by these investigators were almost identical to our values and those reported by Folkow & Halicka (1968). Although the different stimulation patterns have not been compared directly, the present data suggest that both procedures give similar results for peak oxygen consumption, extraction and blood flow in isolated single cat muscles.

There are only few studies in which it has been attempted to relate *in vivo* maximal oxygen consumption rates in muscle to mitochondrial volume (cf. Hoppeler & Lindstedt, 1985). Under conditions of maximal voluntary exercise it is found that 1 cm³ of skeletal muscle mitochondria consumes about 5 ml O₂ min⁻¹ in woodmice (Hoppeler *et al.* 1984), birds (Pennycuik & Rezende, 1985) and humans (Hoppeler & Lindstedt, 1985; data from Saltin, 1985). The present study indicates that the maximal mitochondrial oxygen consumption in isolated cat muscles is somewhat lower than values obtained during voluntary exercise (range in the present study 0.81 to 2.29 ml O₂ cm⁻³ min⁻¹). *In vitro*, the oxygen consumption rates of mitochondria in isolated cardiac myocytes under control and uncoupled conditions are reported to be 2.1 and 4.2 ml O₂ cm⁻³ min⁻¹, respectively (calculated from Jones & Kennedy, 1986, assuming a volume density of 33 % for mitochondria in rat heart cells). Further, Schwerzmann *et al.* (1986) reported isolated liver mitochondria to consume from 0.48 to 3.40 ml O₂ cm⁻³ min⁻¹ depending on substrate. It may be noteworthy that in the latter study the surface density of inner mitochondrial membranes was virtually

identical to the values found in this study. Taken together, the currently available evidence does not exclude that peak blood flow and peak oxygen consumption in surgically isolated autoperfused muscles could be systematically lower than values occurring during maximal voluntary exertion.

The aim of this study was to test the hypothesis that oxygen consumption correlates with volume density of mitochondria when both variables are estimated in identical muscle compartments for conditions of peak oxygen consumption. Our data show such a correlation for muscles which vary naturally in their oxidative potential (soleus and control gracilis) as well as for glycolytic muscles transformed into oxidative muscles (Fig. 3). The correlation between single-muscle $\dot{V}_{O_2, \max}$ and muscle mitochondrial volume indicates that equal amounts of mitochondria consume equal amounts of oxygen, whether these mitochondria are located in oxidative or glycolytic muscles. By the same token, chronic electrical stimulation leads to a proliferation of mitochondria which are structurally and functionally similar to pre-existing mitochondria. The finding that there is a direct proportional relationship between single-muscle $\dot{V}_{O_2, \max}$ and muscle mitochondrial volume does not conflict with reports on disproportionate relative changes of whole-body $\dot{V}_{O_2, \max}$ and muscle oxidative capacity in endurance exercise (Henriksson & Reitman, 1977). Endurance training leads to structural and functional changes which are limited to the muscles used in any particular training regimen (Saltin, Nazar, Costill, Stein, Jansson, Essen & Gollnick, 1976; Rösler, Conley, Claassen, Howald & Hoppeler, 1985). We would suggest that the magnitude of the central, cardiovascular adaptations (reflected as a change in $\dot{V}_{O_2, \max}$) depends on the stress imposed on this system by the energetic requirements of the exercised muscle mass. If this muscle mass is very small (less than 1 % of the total muscle mass in the present experiment) it is likely that very large changes in muscle oxidative capacity will have no measurable effect on a whole-body $\dot{V}_{O_2, \max}$.

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